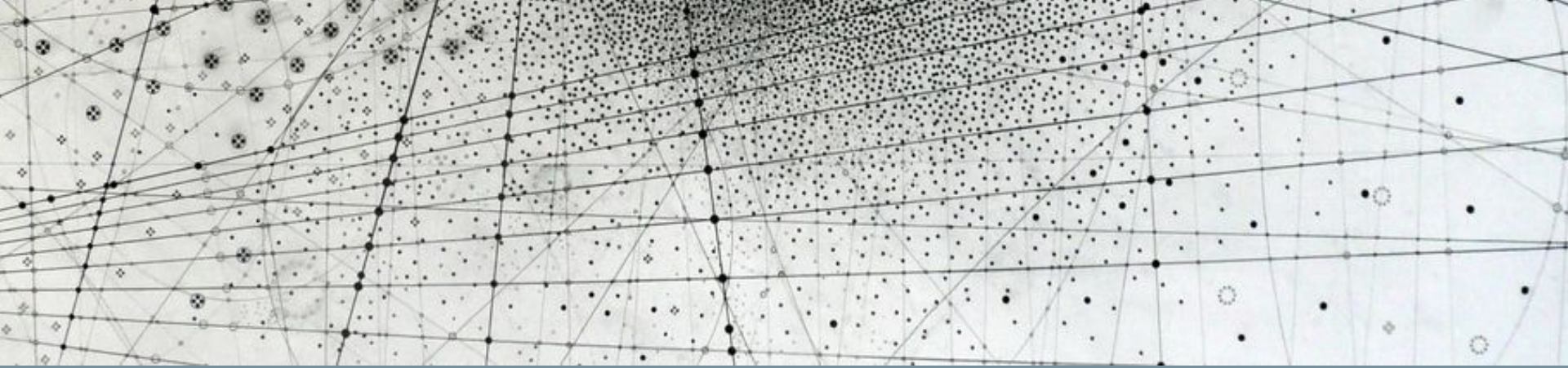




Research Methods in Machine Learning & Physics

Quick recap

Moving toward a unified picture of SSL

All SSL methods are designed to avoid the following scenario:

"Embed these so that
their representations
are similar."



Identical representations
means the loss is 0 😊

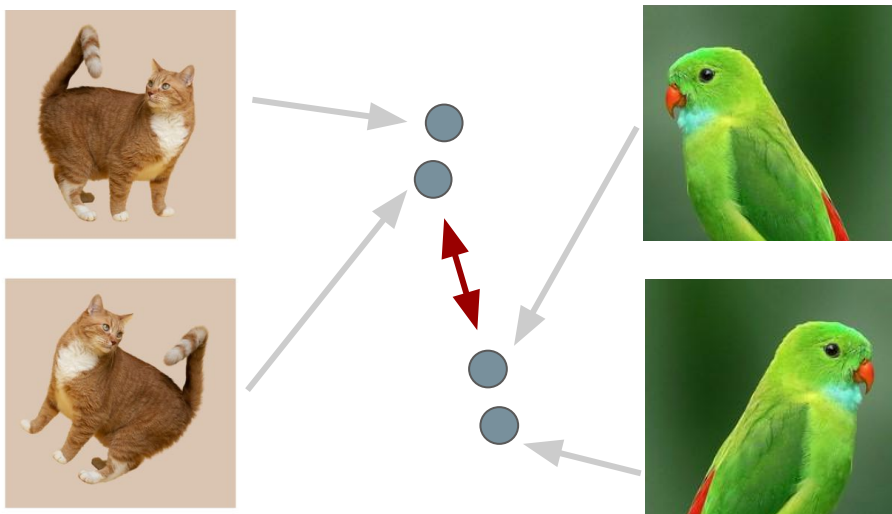
"Now embed these so
that their representations
are similar."



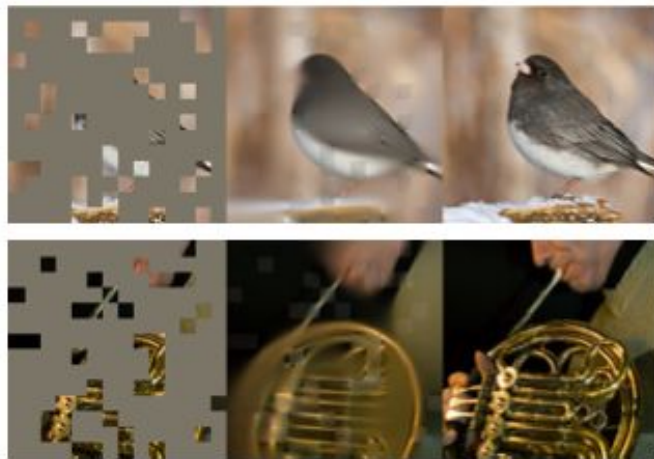
Moving toward a unified picture of SSL

How can we avoid learning a trivial solution? → **Need to enforce some variation!**

Contrastive methods use **negative pairs** to push apart various representations:



Generative methods use **masking**:



Note: only possible if you have a discrete input space!

Non-contrastive SSL methods

Masked modeling

- In the input (e.g. pixel) space or the latent space (JEPA)

Apply statistical constraints ("redundancy reduction")

- VICReg
- Barlow Twins

Use an asymmetric architecture ("self-distillation")

- BYOL
- DINO

Unified theories of self-supervised learning

6 CONCLUSION

Through an analysis of their criteria, we were able to show that sample-contrastive and dimension-contrastive methods have learning objectives that are closely related, as they are effectively minimizing criteria that are equivalent up to row and column normalization of the embedding matrix. This suggests a certain duality in the behavior of such methods, which we studied empirically. Through the lens of variations of VICReg, we were able to study popular design choices in self-supervised loss functions and show their lack of impact on performance, significantly improving the robustness to embedding dimension of VICReg along the way. Motivated by our theoretical findings, we performed ample hyperparameter tuning on SimCLR and were able to close its performance gap with VICReg. We also showed that the normalization strategy does not play an important role in performance. This further reinforces the similarities between methods as predicted by our theoretical results. We expect that our results will help extend theoretical works in self-supervised learning to a wider family of methods, as well as help analyses by deriving criteria that are easier to work with. We also expect that our findings will help alleviate preconceived ideas on contrastive and non-contrastive learning. If one thing must be remembered from this work, it is that *dimension-contrastive and sample-contrastive methods are two sides of the same coin*. Finally, perhaps the most important message of this work is to show that different SOTA SSL methods can be unified.



Multimodal models

A **modality** is a specific way of perceiving the world –
it's a concept deeply rooted in our own five senses.

(In ML, common modalities include text, audio, video, etc.)

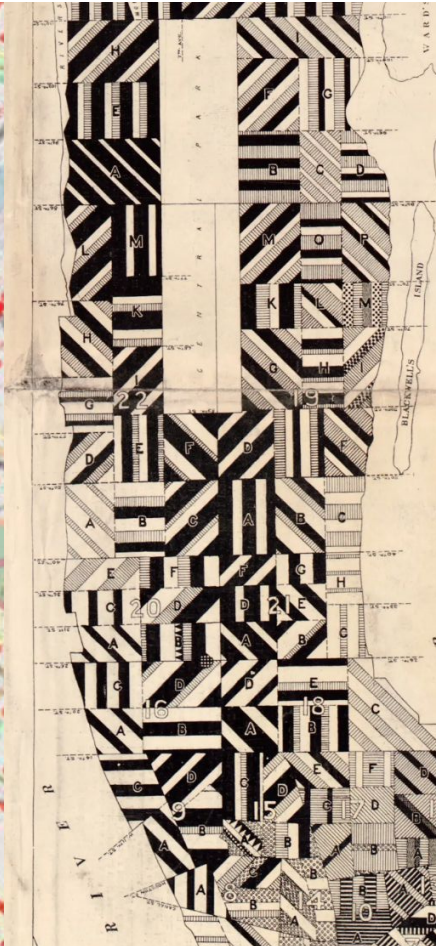




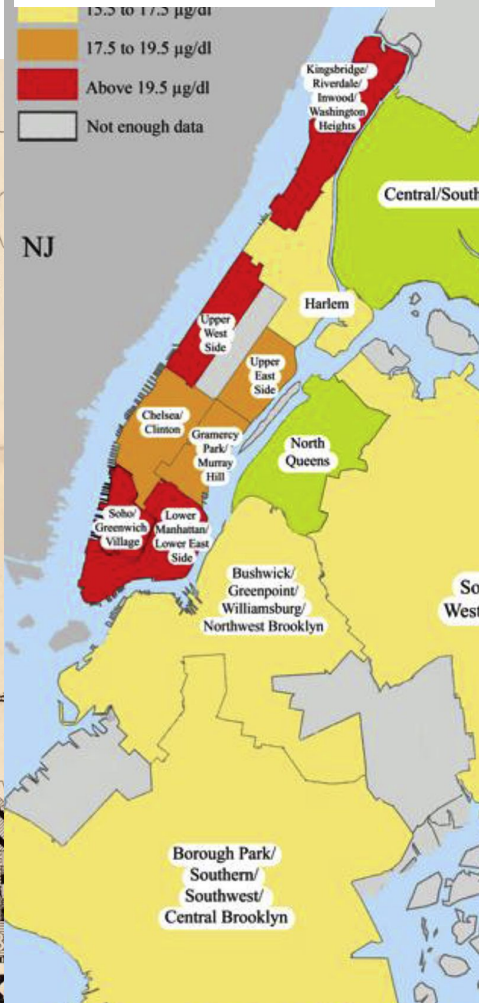
Top-rated pizza places



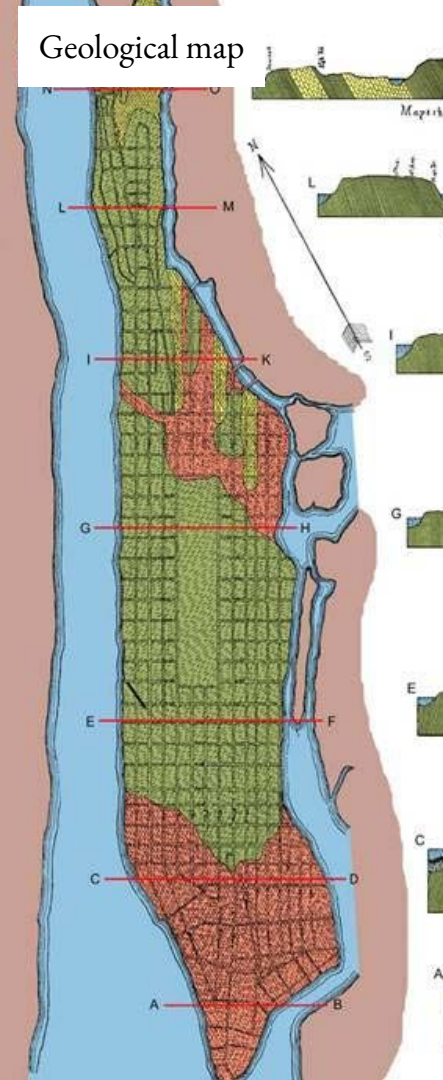
1894 map of immigrant populations in Manhattan



Lead levels found in pigeons



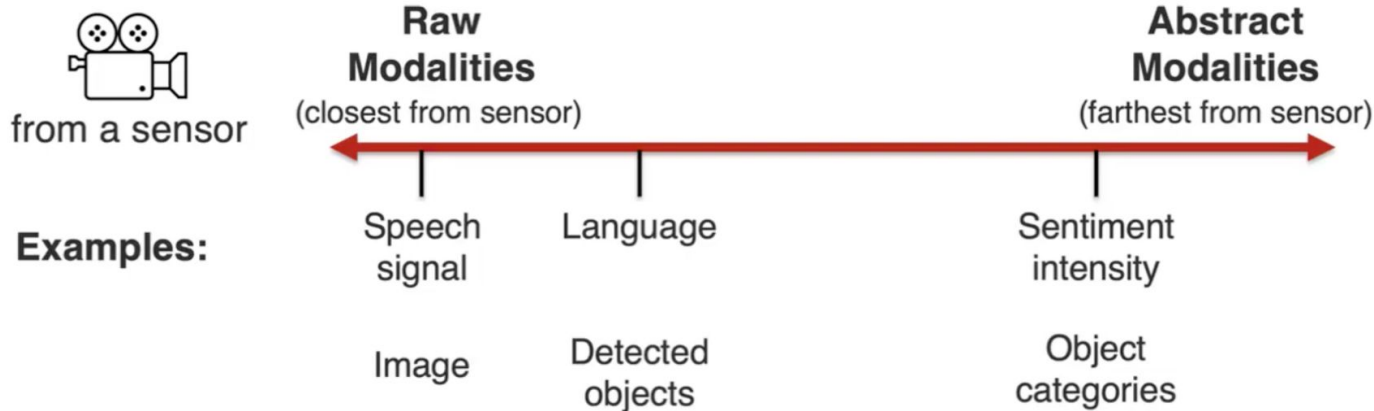
Geological map



Alternate definition:

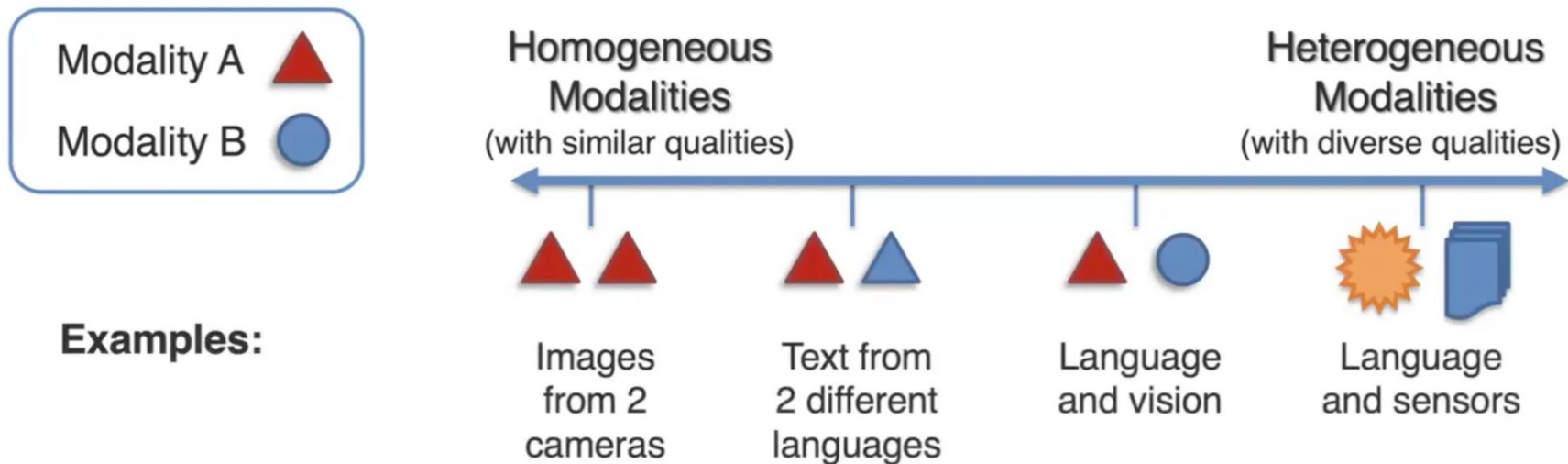
Modality

Modality refers to the way in which something expressed or perceived.

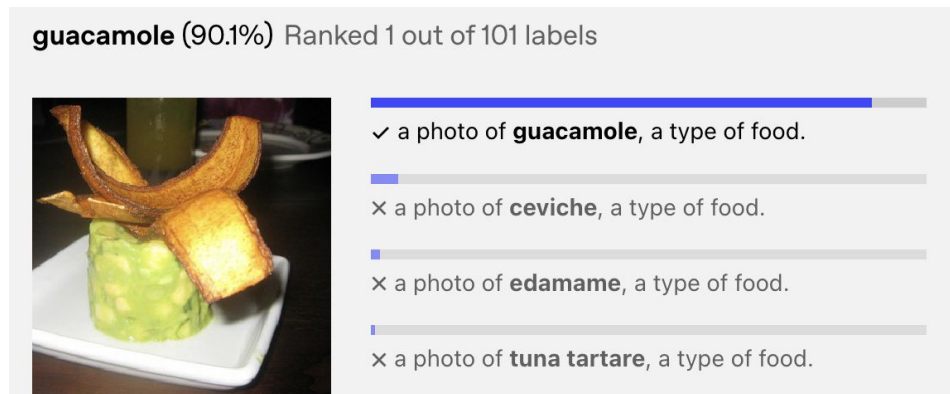


Multimodal ML involves data analysis using **heterogeneous & interconnected** data sources.

Homogeneous vs. heterogeneous modalities:



AI industry has multimodal models, but the modalities are relatively few and well-defined:

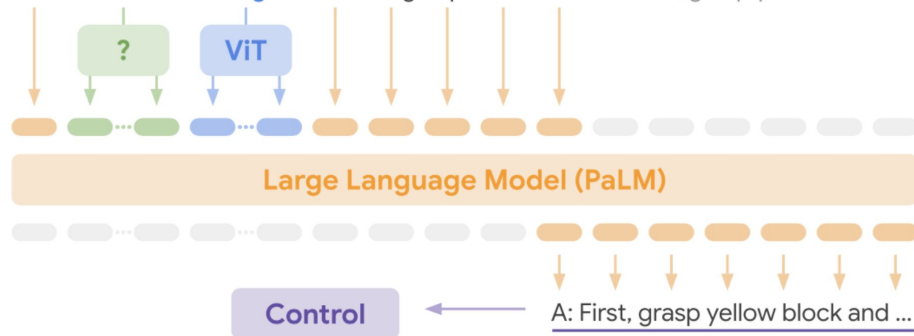


Text & images

(OpenAI's CLIP, 2021)

PaLM-E: An Embodied Multimodal Language Model

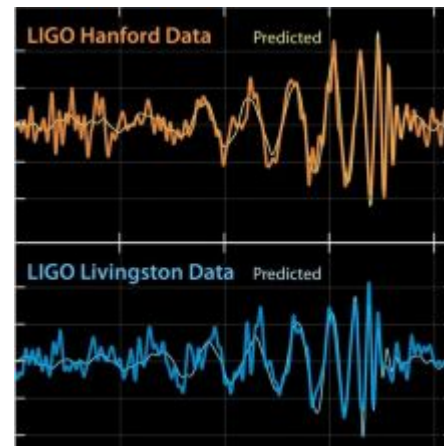
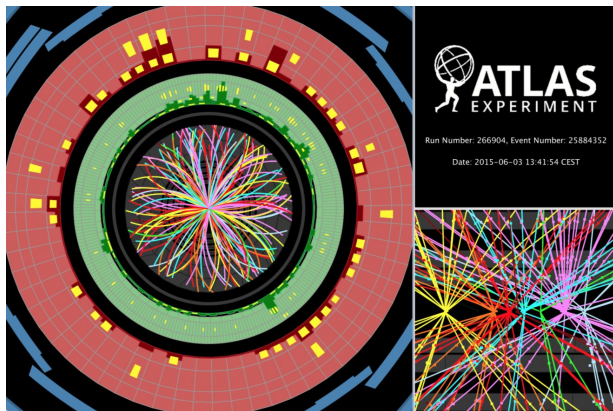
Given `<emb>` ... `<img>` Q: How to grasp blue block? A: First, grasp yellow block



Text & images &
robotic instructions

(Driess et. al., [PaLM-E](#), 2023)

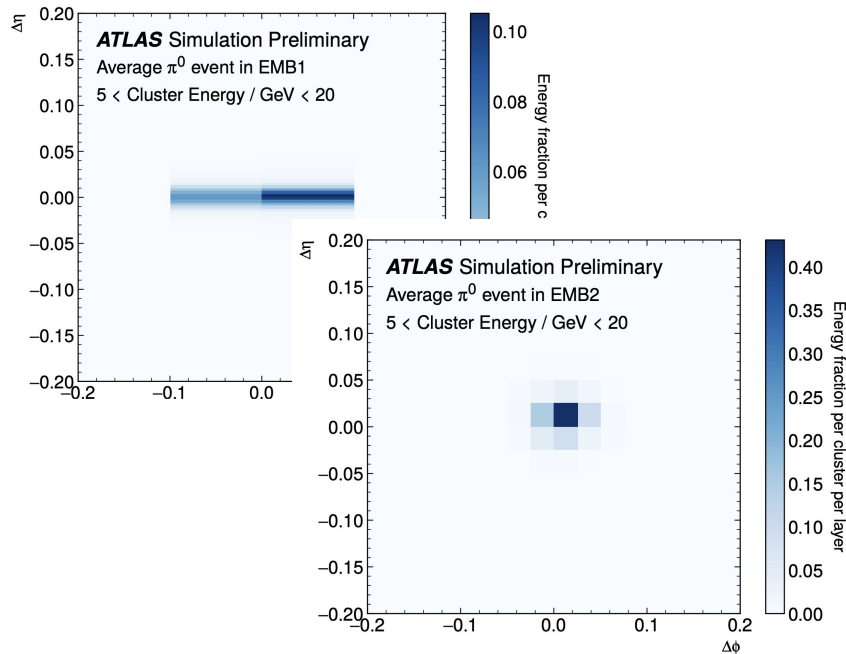
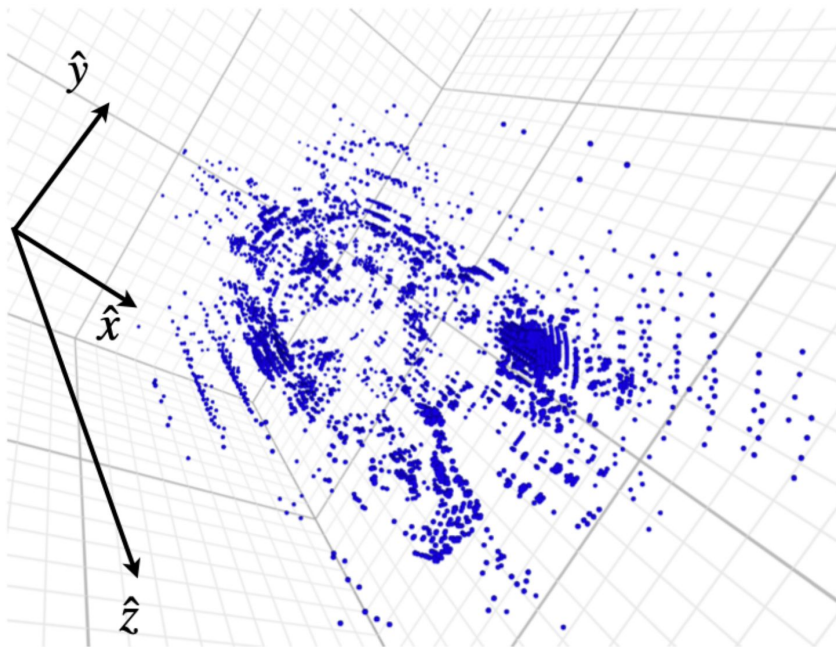
Physics data, on the other hand, is highly diverse and multimodal.



Common embedding?

Physics data, on the other hand, is highly diverse and multimodal.

e.g. in a high-energy particle physics detector, one might measure a single collision event with several layers of calorimeter images, point clouds, and even metadata.



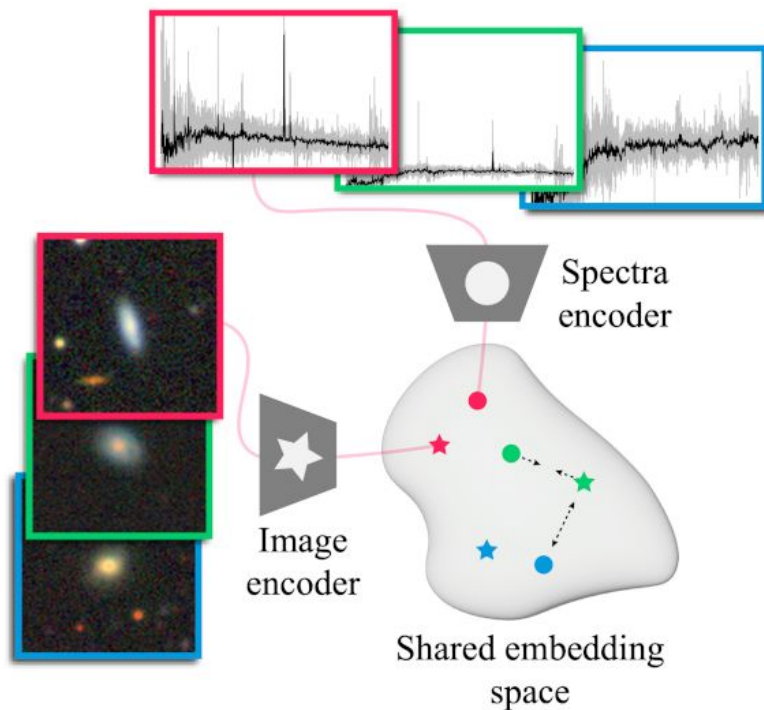
Multimodal models see promising performance in industry and in science.

Significant evidence suggests that **including multiple views/modalities** in a single model can improve performance on downstream tasks relative to unimodal models:

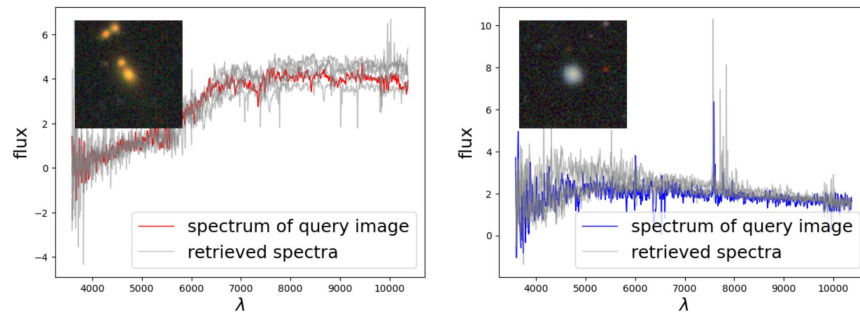
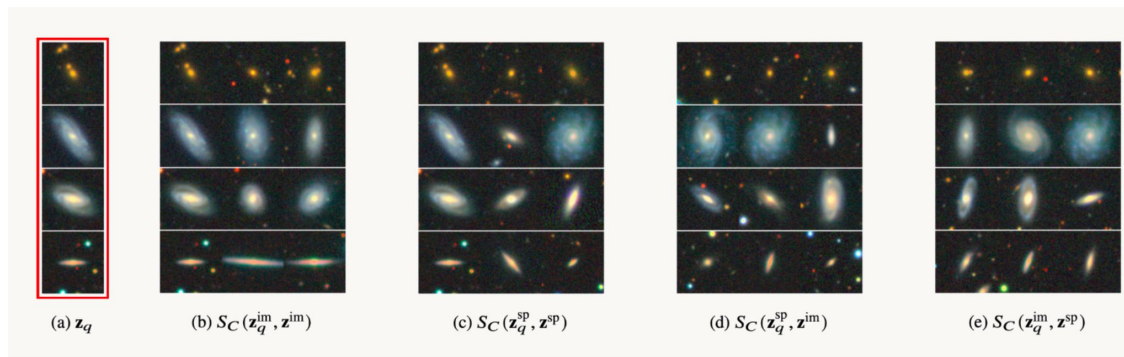
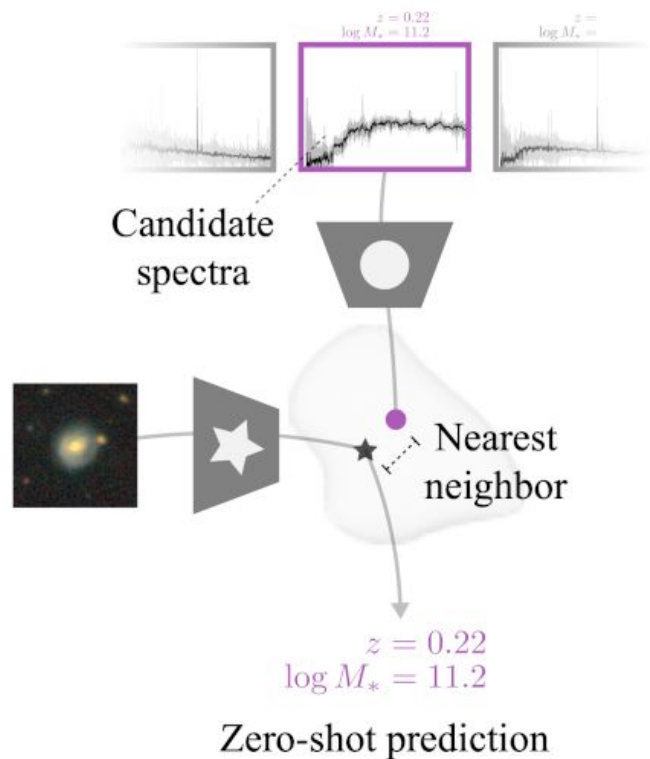
	Dataset Examples	ImageNet ResNet101	Zero-Shot CLIP	Δ Score
ImageNet		76.2	76.2	0%
ImageNetV2		64.3	70.1	+5.8%
ImageNet-R		37.7	88.9	+51.2%
ObjectNet		32.6	72.3	+39.7%
ImageNet Sketch		25.2	60.2	+35.0%
ImageNet-A		2.7	77.1	+74.4%

(CLIP, 2021)

Multimodal models see promising performance in industry and in science.



The resulting latent space is semantically meaningful and can be used for cross-modal retrieval:



(f) $S_C(\mathbf{z}_q^{\text{im}}, \mathbf{z}^{\text{im}})$

×

Search a galaxy

RA

?

Dec

?

190.045

62.883

Search from

?

☒ Image
☐ Spectrum
☐ Image+Spectrum

Compare with

?

☒ Image
☐ Spectrum
☐ Image+Spectrum

Search

Random (bright)

Random

Number of similar galaxies to display

9

9

25

? Instructions

▼

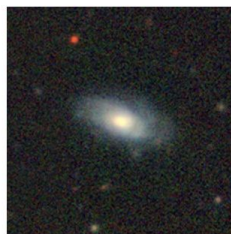
Built with Streamlit

❤

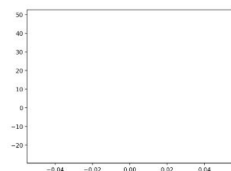
Welcome to AstroCLIP Galaxy Finder

Check out the retrieval abilities of our new multimodal model with this interactive web app! Read our [blog post](#) to learn more.

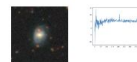
Query:



RA, Dec = (190.0454, 62.8834)



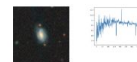
Most similar objects



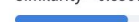
Similarity = 0.8602



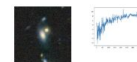
RA, Dec =
(252.9043,
36.1280)



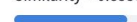
Similarity = 0.8567



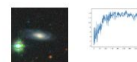
RA, Dec =
(242.9475,
55.7267)



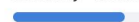
Similarity = 0.8550



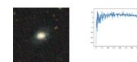
RA, Dec =
(235.5068,
44.5857)



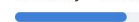
Similarity = 0.8456



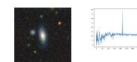
RA, Dec =
(148.3912,
37.3902)



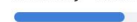
Similarity = 0.8447



RA, Dec =
(177.1778,
52.1176)



Similarity = 0.8387

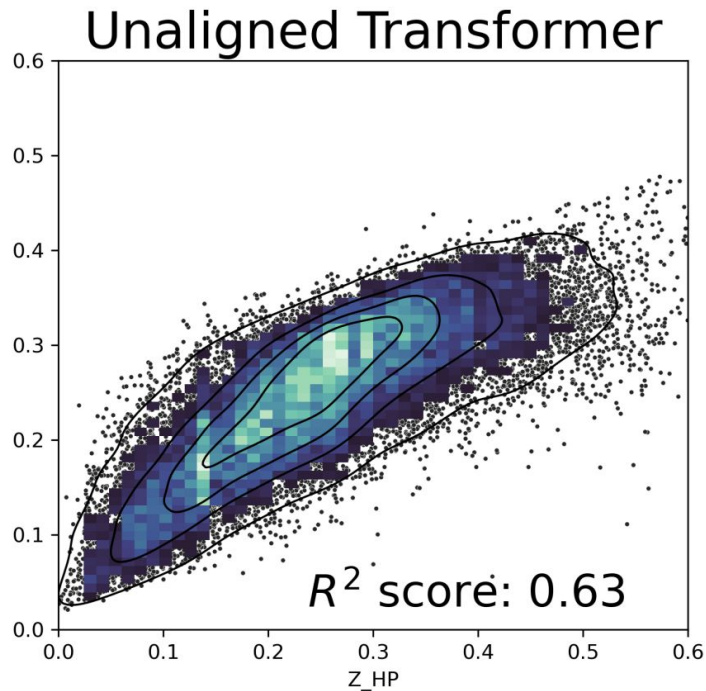
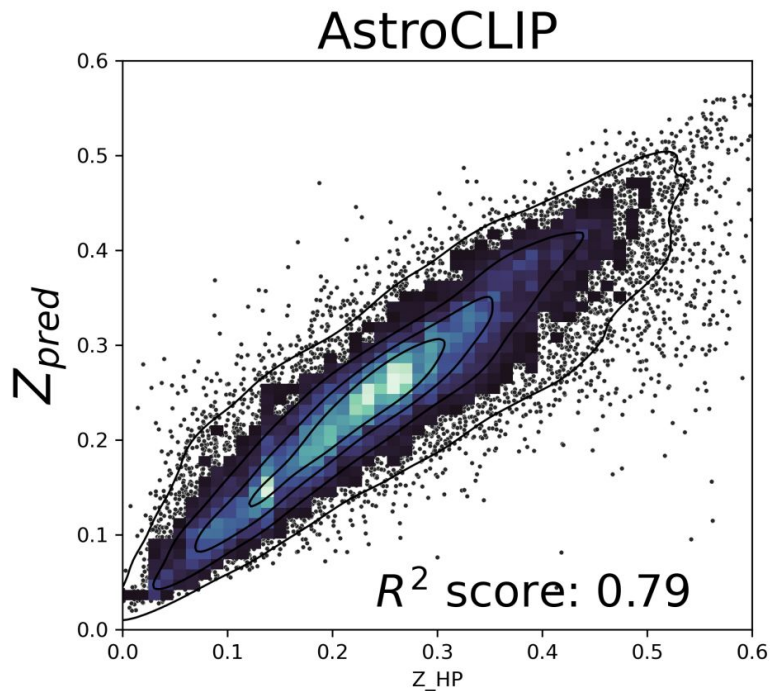


RA, Dec =
(188.5688,
55.2721)

Fullscreen

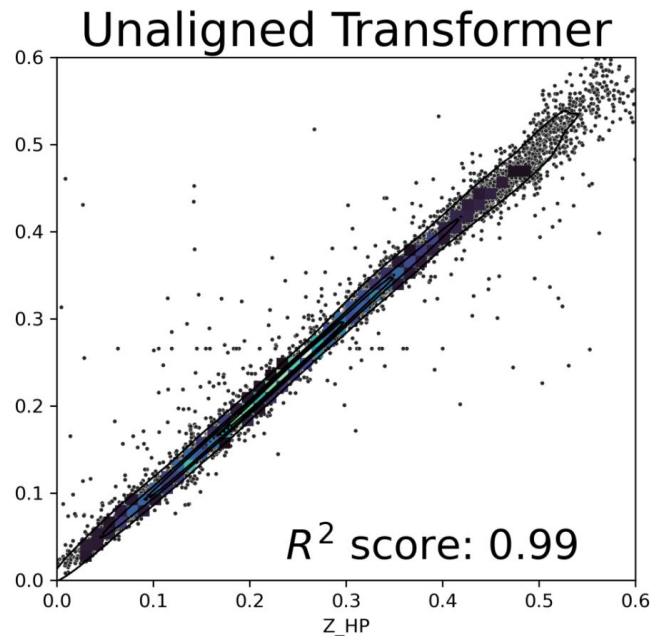
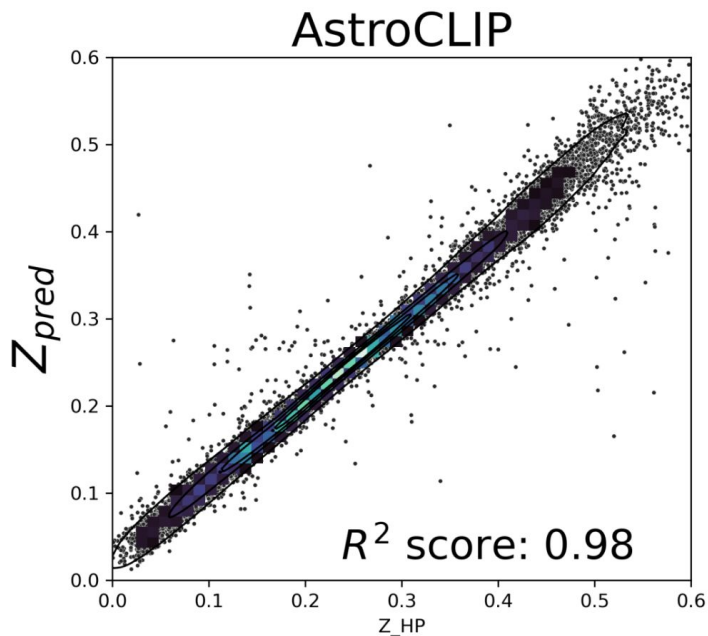
The learned joint embeddings are also useful for making better downstream predictions, e.g. redshift:

AstroCLIP is able to better estimate the redshift from the galaxy image alone in a zero-shot setting compared with models trained only on galaxy images:



The learned joint embeddings are also useful for making better downstream predictions, e.g. redshift:

And the joint embedding also doesn't lose much, if any, information, compared with the pure embeddings of spectra alone:



In-class work

- Breakout room 1: **Physics data modalities**
 - Come up with a list of as many modalities as possible within physics
 - Provide concrete examples for each
 - If you have time, organize these from raw → abstract, or by mutual info
- Breakout room 2: **CLIP code**
 - Explore the CLIP colab notebook and prepare a 3-min walkthrough
- Breakout room 3: **AstroM³ paper**
 - Summarize the high-level strategy and results in 1-2 slides

We'll reconvene at the top of the hour to share our highlights!